8. Resonance

Swings, wine glasses and the LC tank

Every circuit with an L and a C has one note it likes

Lesson 8 · about 90 minutes · everyone builds; battery power only

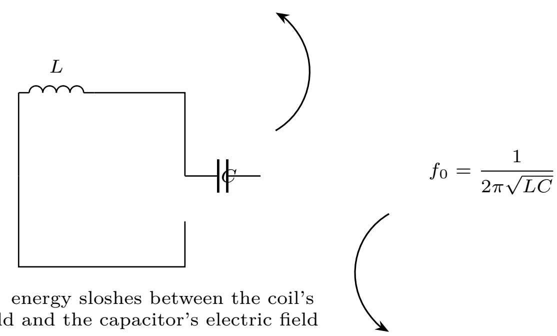
What you need

- String and a weight, for a pendulum
- An inductor: the 100-turn coil from Lesson 4, or a 1 mH choke
- Capacitors: 0.01 μF, 0.1 μF, 1 μF (film, not electrolytic)
- A signal generator, or a phone app with a tone generator and an amplifier
- An oscilloscope if you have one; otherwise a multimeter on AC volts
- A wine glass and water

Predict first

You push a child on a swing. Why does pushing at exactly the right moment work so much better than pushing hard at random moments? Write the answer in your own words. Then: an LC circuit rings at one frequency. If you *quadruple* the capacitance, does the frequency double, halve, quarter, or something else?

Do it



an LC tank: energy sloshes between the coil's magnetic field and the capacitor's electric field

1. **The pendulum.** Hang a weight on a string. Push it at random — it goes nowhere. Push it once each time it swings back to you — it climbs quickly. Now shorten the string by half and find the new rhythm. Every oscillator has one frequency at which small, well-timed pushes accumulate.
2. **The wine glass.** Wet a finger and run it round the rim until it sings. Add water and do it again — the note drops. You changed the effective mass of the oscillator, and the frequency followed.
3. **Build the tank.** Connect your coil and a 0.1 μF capacitor in parallel. Drive it from the signal generator through a 1 kΩ resistor and measure the AC voltage across the tank.
4. **Sweep it.** Step the frequency up in stages, recording the voltage each time. Plot voltage against frequency. There is one frequency where the reading peaks sharply — that is f_0.
5. **Change one thing.** Swap to 0.01 μF and find the peak again. Then 1 μF. Compare with your prediction: a hundred-fold change in C should move f_0 by a factor of ten, because the formula has a square root in it.
6. **Check the formula.** Compute $1/(2\pi\sqrt{LC})$ for each and compare with what you measured.

What it means

Energy sloshes	Charge the capacitor and let it go: it drives current through the coil, which builds a magnetic field. When the capacitor is empty the field collapses, and by Lesson 5 that collapse drives current onward, recharging the capacitor the other way round. Back and forth — the exact electrical analogue of the pendulum.
The resonant frequency	$f_0 = 1/(2\pi\sqrt{LC})$. It depends only on the coil and the capacitor, not on how hard you hit it.
Why the peak is sharp	At f_0 small pushes arrive in step and add up, so the voltage in the tank climbs far above the driving voltage. Off resonance they arrive out of step and cancel.
Q	How sharp that peak is, is called the Q of the circuit. Low resistance means high Q, means a taller and narrower peak, means more energy accumulated before losses eat it. A Tesla coil secondary wants the highest Q you can build.
Damping	Add a resistor in series and sweep again. The peak flattens. Resistance is what stops the swing.

The link to the coil

Both halves of a Tesla coil are LC tanks. The primary is a few turns of heavy conductor plus the tank capacitor. The secondary is a thousand turns of fine wire, and its capacitance is not a component at all — it is the topload and the coil's own self-capacitance against the surrounding air. **Tuning a Tesla coil means adjusting one tank until its f_0 matches the other's**, and you do it by moving the primary's tap, which changes L. Every hour you will later spend tuning the coil is this lesson.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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